Values variables and types

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Python readiness portfolio

TEACHER GUIDE · 1 OF 9

Prepare the purpose, prerequisites and materials

> Prepare Values variables and types: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Distinguish integer, floating-point, text and Boolean **value**s

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

From a paper algorithm to Python

Without opening the old lesson, explain one key idea from From a paper algorithm to Python. Give an example and a mistake that the idea helps you avoid.

Look for: Translate a short paper calculation into a Python script

Reference example, not the only acceptable wording: packs = 4; per_pack = 6; total = packs * per_pack; print(total), each on its own line. The first two bind quantities, the third computes 24 and the fourth displays 24.

Assignment alone produces no displayed **output** in this script.

Focus on scalar **value**s and rebinding; lists and mutation come later. Avoid saying every assignment makes a deep copy. For the integers used here, rebinding one name leaves the other bound to its prior **value**.

Reference: Python tutorial introduction, https://docs.python.org/3/tutorial/introduction.html, and built-in **type**s, https://docs.python.org/3/library/std**type**s.html.

Outcomes

Distinguish integer, floating-point, text and Boolean **value**s

Trace **reassignment** without treating a name as a live formula

Choose a suitable representation for a count and an identifier

Preparation

Python 3 editor prepared in Week 1 and fresh **value**s.py script

Name cards and **value** cards marked with int, float, str or bool

Fictional shelf labels and a blank trace table

Prerequisite answer

It displays the current **value** 24. Assignment and display are separate operations.

End goal for the pictorial case

Trace **value**s **variable**s and **type**s and explain the changed program before running it.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain and model the mechanism

> Model Values variables and types using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Explain Values variables and types · 1

A count and a shelf label can look similar while needing different operations. The integer 8 is a number; "08" is text that preserves a leading zero.

You can count eight objects, but adding two shelf labels is usually meaningless. This week you will examine **value**s, trace changes to names and explain which representation a task needs.

Explain Values variables and types · 2

Use the saved Python 3 script workflow from Week 1. Make a new file **value**s.py, predict before running and keep actual **output** separately. Use fictional labels only.

Four useful **type**s are int for whole numbers, float for floating-point numbers, str for text, and bool for True or False. A **type** describes a **value**, not whether its meaning is sensible for the task.

type(count) produces the **type** of the **value** bound to count. These print lines display <class 'int'> and <class 'str'>. The quotes in "08" mark text; they are not displayed by print(shelf).

True and False have capital initials and no quotes. "False" is text, not a Boolean.

Term	Meaning
Value	A piece of data such as 8 or a text label
Variable	A name bound to a value during execution
Type	A category of value that determines available operations
String	A sequence of text characters
Reassignment	Binding an existing name to a new value

A saved value is not a live link

INSTRUCTION	COUNT NOW	SAVED NOW
count = 8	8	Not assigned
saved = count	8	8
count = count + 2	10	8

Evaluate the right side using the current value, then bind the left-hand name. The saved integer does not update when count is reassigned.

Explain Values variables and types · 3

The new saved = count binds saved to the current 10, so print(saved) displays 10. Without that assignment it stays 8.

Exit: turning "08" into 8 loses the leading zero; identifier addition has no defined meaning in the stated task. -3 is an integer but cannot be a nonnegative count.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

A saved value is not a live link

INSTRUCTION	COUNT NOW	SAVED NOW
count = 8	8	Not assigned
saved = count	8	8
count = count + 2	10	8

Evaluate the right side using the current value, then bind the left-hand name. The saved integer does not update when count is reassigned.

```
count = 8
saved = count
count = count + 2
print(count)
print(saved)
```

The **output**s are 10 and 8 on separate lines. saved = count uses the **value** available at that moment.

It does not store a formula that keeps following count. count = count + 2 is valid assignment: read the old 8, calculate 10, then rebind count. It is not a claim that 8 equals 10.

> **After count becomes 10, add saved = count and print(saved). Predict the new output and explain why this is different from expecting saved to change automatically.**

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Values variables and types

A number and its printed-looking text. A program stores a numeric badge count and a **string** that looks like that count.

The facts we will use

> **Supplied facts**
>
> Python 3: count=8; tag="8"; print(count+2); print(tag+"2").

AI ACADEMY / PYTHON-BRIDGE/2

Values variables and types

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Inspect types

count is an integer; tag is a string.





2. Apply addition

Integer addition gives 10; string concatenation gives the text 82.





3. Predict a mixed operation

count+tag raises TypeError rather than silently converting tag.

CHANGE ONE THING

Change count to 10, leaving tag unchanged.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Inspect types	count is an integer; tag is a string.
2. Apply addition	Integer addition gives 10; string concatenation gives the text 82.
3. Predict a mixed operation	count+tag raises TypeError rather than silently converting tag.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Values carry **type**s, not merely appearances. The plus operator performs numeric addition for these integers and concatenation for these **string**s; the mixed operation is invalid here.

What this example does not establish

Keep the stated limit

These examples use built-in integers and **string**s; operator behaviour depends on **type**s and is not proof of general model reasoning.

Change one thing in the pictured case

Change one thing

Change: Change count to 10, leaving tag unchanged.

Predicted result

Prediction: The two valid **output**s become 12 and 82.

Why the result follows

Reason: Only the integer **variable** changes; the **string** still contains the character 8.

Ask while showing the picture

Ask while showing the picture · 4

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Values variables and types, reduce support deliberately and check individual reasoning before discussing answers.

Choose the pacing route

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve assignment versus print and collect the prerequisite.
7 to 20 minutes	Compare count and shelf label; model type output and complete W1.
20 to 33 minutes	Use the diagram and move name cards while tracing the count program.
33 to 45 minutes	Complete W2 with predicted and actual outputs in separate columns.
Session 2 start to 8 minutes	Retrieve the two saved values and explain the extra assignment in the guided task.
8 to 20 minutes	Run the numeric and text operations; use W3 to diagnose a TypeError.
20 to 35 minutes	Collect W4 independently before support, including the seat-label choice.
35 to 45 minutes	Complete W5 and the exit; separate type from permitted value.

Use the worked example in Sections 3 and 6. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Use movable name cards and immutable number cards. Keep quotations visible on text cards and show source beside **output**. Offer an oral trace but record assistance.

Extension: explain why a **type**-correct number such as -3 is not acceptable as the number of seats.

Reduce help deliberately

Completion practice

Completion task: Trace the five-line count and saved program. Record both printed **output**s, then predict the result of assigning saved = count after the change. Check with a run or label a paper trace.

Reasoning to listen for

Teacher check: Original **output**s 10 then 8. After the new assignment, saved becomes 10. Keep actual **observation**s separate. Reading an integer **value** earlier does not create a live link to another name.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for **Value**s **variable**s and **type**s. Proposed explanation: “Digits always mean a number”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: Quotes make a **string**, even when every character is a digit.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

> Predict both print **output**s and explain why they differ. Does count+tag work, and what happens to each valid **output** after count becomes ten?

Reasoned teacher answer

The **output**s are 10 and 82. The first adds integers; the second joins **string**s. count+tag raises **Type**Error. Changing count to 10 yields 12, while tag+"2" remains 82.

Changed-case reasoning: The two valid **output**s become 12 and 82. Only the integer **variable** changes; the **string** still contains the character 8.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the investigation and diagnose misconceptions

> Run the Values variables and types investigation and use the first unsupported step to choose a teaching response.

Digits always mean a number: Quotes make a **string**, even when every character is a digit.

Changing one scalar name updates every earlier assignment: Trace the **value** bound at each assignment; no formula link is stored.

The right **type** guarantees valid data: A negative integer can be an invalid object count.

Lowercase false is Python's Boolean false **value**: The literal is False; lowercase false is a name and is undefined in a fresh script.

Use Section 9 for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Values variables and types questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Guided and exit questions

The new saved = count binds saved to the current 10, so print(saved) displays 10. Without that assignment it stays 8.

Exit: turning "08" into 8 loses the leading zero; identifier addition has no defined meaning in the stated task. -3 is an integer but cannot be a nonnegative count.

W1 Identify the value and its type

Workbook question · W1

Classify 12, 1.5, "12", False and "False". Explain why the last two are different.

Teacher reasoning · W1

12 is int; 1.5 is float; "12" is str; False is bool; "False" is str. Quotes mark text even when the characters spell a Boolean name.

W2 Trace a saved value

Workbook question · W2

Trace the five-line count and saved program. Record both printed **output**s, then predict the result of assigning saved = count after the change. Check with a run or label a paper trace.

Teacher reasoning · W2

Original **output**s 10 then 8. After the new assignment, saved becomes 10. Keep actual **observation**s separate. Reading an integer **value** earlier does not create a live link to another name.

W3 Diagnose the operation

Workbook question · W3

Predict 6 + 3, "6" + "3" and "6" * 3. Explain what happens for "6" + 3. Which expression counts six groups of three?

Teacher reasoning · W3

9, text 63, text 666, and **Type**Error respectively. Six groups of three needs numeric multiplication 6 * 3, giving 18; none of the listed successful expressions calculates that total.

W4 Trace a new pair of names independently

Workbook question · W4

A fresh script runs seats = 12; booked = seats; seats = seats - 5; print(seats); print(booked), with each statement on a separate line. Predict both **output**s. Then add booked = seats + 1 and print(booked).

Explain the evaluation order and choose a **type** for a seat label "07".

Teacher reasoning · W4

The **output**s are 7 and 12, then 8 after the added assignment. The right side reads the current seats **value** 7 before binding booked to 8. The seat label remains str to preserve its leading zero.

W5 Separate type from validity

Workbook question · W5

Give a suitable **type** for a counter count, a length of 2.5 metres, a shelf ID with a leading zero, and a yes-or-no availability flag. Give a count **value** with the right **type** that should still be rejected.

Teacher reasoning · W5

Use int, float, str and bool respectively. A negative integer such as -3 has **type** int but is invalid for a nonnegative physical count. **Type** checking and checking the task's permitted **value**s are different.

Independent assessment and fresh retry

Teacher key · Independent assessment and fresh retry

W4 needs **output**s 7, 12 and 8 with a right-side-first explanation and str for "07". After support, retry with stock = 9, spare = stock, stock = stock - 4: printing stock and spare gives 5 and 9; spare = stock + 2 then gives 7.

Record independent reasoning before giving the retry.

Assess independent transfer

Independent question · withhold the key

Selected case: Predict both **output** lines and explain why estimate stays unchanged when count is reassigned. Repeat the trace with initial count=0. Explain why note is text even though its printed characters look like a number.

```
count = 4
note = "4"
```

```
estimate = count / 2
fits = count <= 5
count = count + 3
print(count, note, estimate, fits)
print(type(count).__name__, type(note).__name__, type(estimate).__name__,
type(fits).__name__)
```

Selected case answer

> **Teacher key · Selected case answer**
>
> Original **output**s: 7 4 2.0 True, then int str float bool. Estimate was calculated as 2.0 before **reassignment**; it is not a live formula. With initial count=0: 3 4 0.0 True, with the same **type**s.
>
> Note is the quoted **string** "4", independent of the integer count.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

> **Teacher key · Selected case answer**
>
> Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.
>
> Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

TEACHER GUIDE · 6 OF 9

Connect the project, support and next lesson

> Connect Values variables and types to Python readiness portfolio, adapt support and plan a later changed-case check.

Add a labelled **value**s **variable**s and **type**s **evidence** sheet to Python readiness portfolio. Show how you can distinguish integer, floating-point, text and boolean **value**s.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Use movable name cards and immutable number cards. Keep quotations visible on text cards and show source beside **output**. Offer an oral trace but record assistance.

Extension: explain why a **type**-correct number such as -3 is not acceptable as the number of seats.

Week 3 combines numeric **value**s in expressions and uses parentheses to make grouping explicit.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

Plan the delayed return

Suggested return points in this level: python-bridge/3, python-bridge/5, python-bridge/8. Use actual classroom dates, adjusting for holidays and gaps.

One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For Python readiness before the Responsible Plant Classifier, retain a trace of the **value**s, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Values variables and types lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Distinguish integer, floating-point, text and Boolean **value**s

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: Trace the five-line count and saved program. Record both printed **output**s, then predict the result of assigning saved = count after the change.

Check with a run or label a paper trace.

Supported practice answer

Reasoned practice answer: Original **output**s 10 then 8. After the new assignment, saved becomes 10. Keep actual **observation**s separate. Reading an integer **value** earlier does not create a live link to another name.

Check every learner before discussion: Use the worked case for **Value**s **variable**s and **type**s. Proposed explanation: “Digits always mean a number”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Quotes make a **string**, even when every character is a digit.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Change count to 10, leaving tag unchanged. Require a **prediction** and a reason before comparing. Expected result: The two valid **output**s become 12 and 82.

Reason: Only the integer **variable** changes; the **string** still contains the character 8.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the

new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Python readiness portfolio. Keep runnable code, **prediction**s, actual **output**s, one repaired error and an explanation of a function on changed **input**s.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

TEACHER GUIDE · 8 OF 9

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

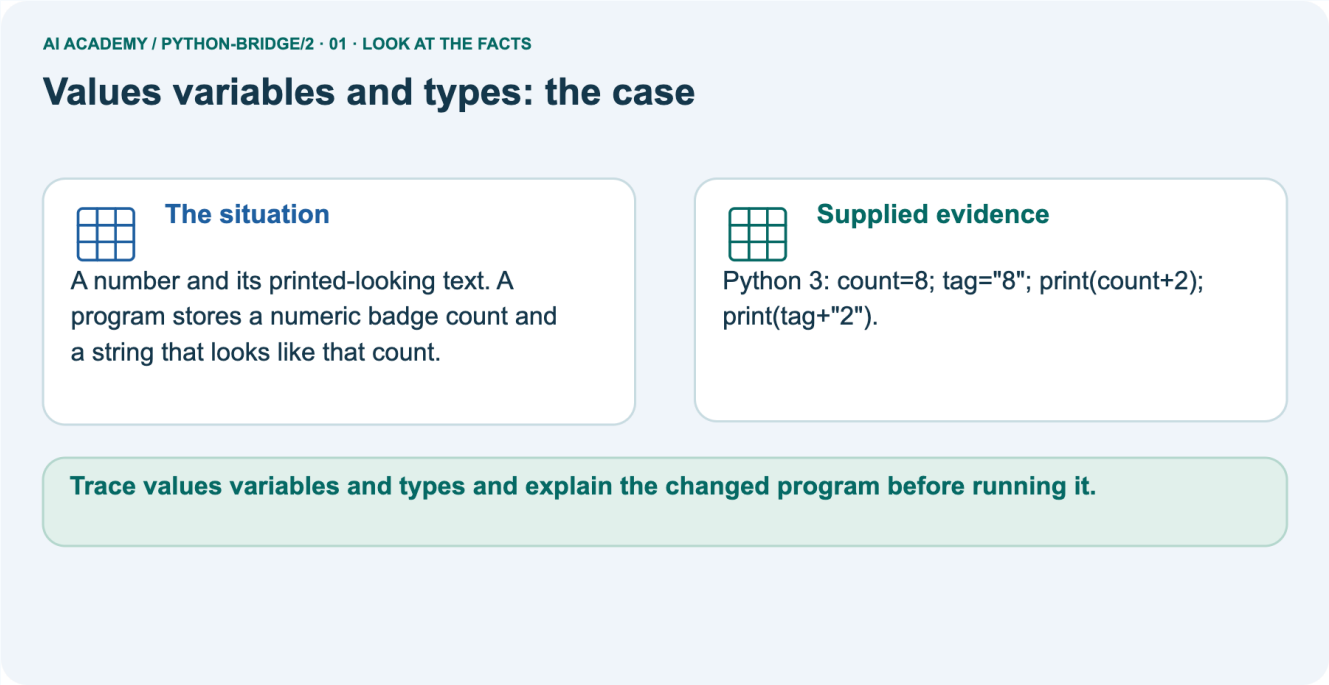
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Values variables and types graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Values variables and types: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

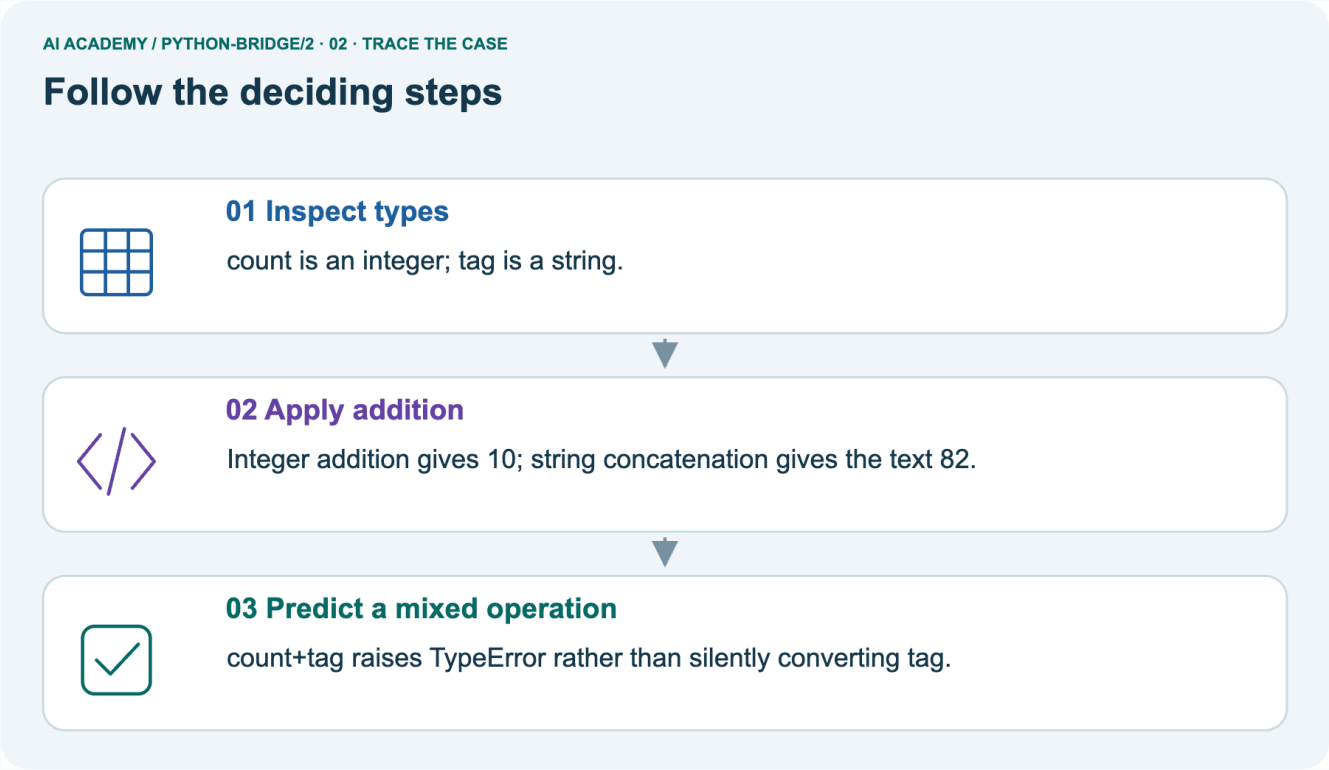
A number and its printed-looking text. A program stores a numeric badge count and a **string** that looks like that count. Supplied facts: Python 3: count=8; tag="8"; print(count+2); print(tag+"2").

Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Inspect **type**s: count is an integer; tag is a **string**. → Apply addition: Integer addition gives 10; **string** concatenation gives the text 82. → Predict a mixed operation: count+tag raises **Type**Error rather than silently converting tag.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. A saved value is not a live link

AI ACADEMY / PYTHON-BRIDGE/2 · 03 · INSPECT THE LESSON MODEL

A saved value is not a live link

Instruction	count now	saved now
count = 8	8	Not assigned
saved = count	8	8
count = count + 2	10	8

Evaluate the right side using the current value, then bind the left-hand name. The saved integer does not update when count is reassigned.

Follow the labelled picture with the matching explanation. Open the vector diagram

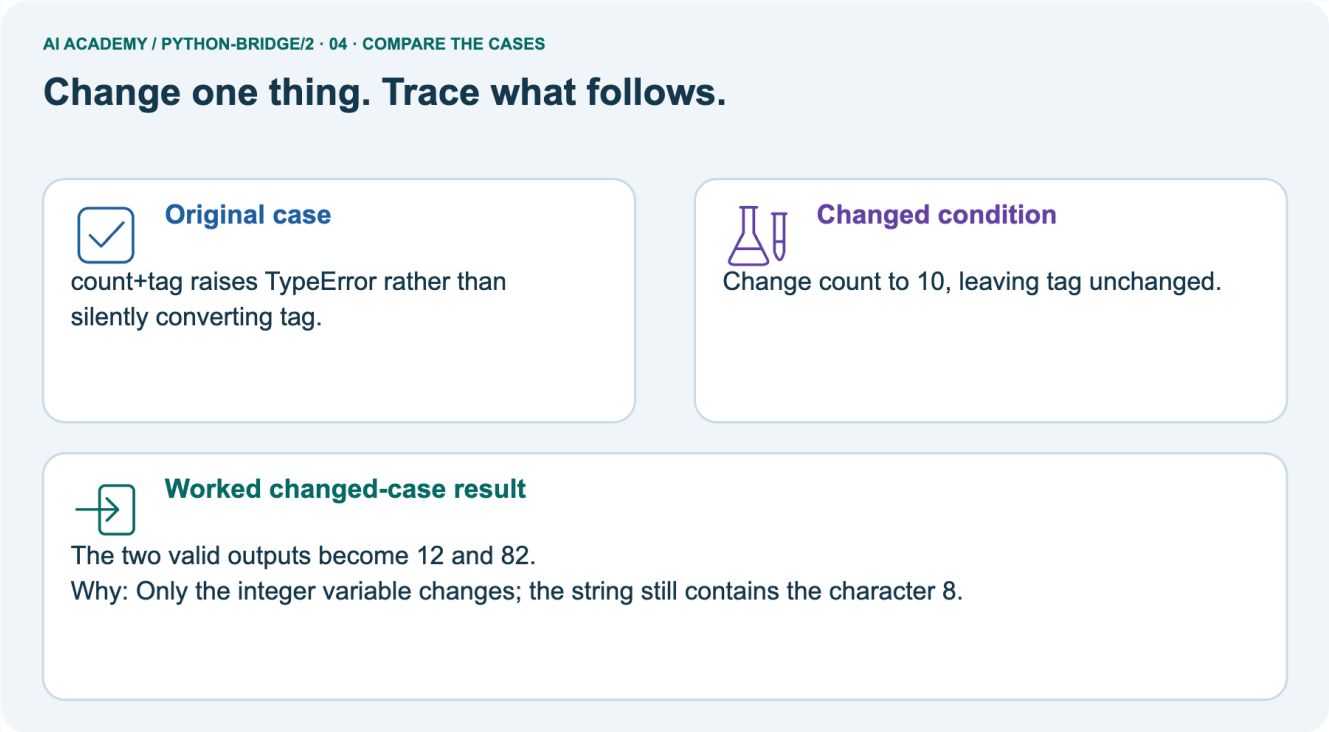
A saved **value** is not a live link. count = 8 | 8 | Not assigned. saved = count | 8 | 8. count = count + 2 | 10 | 8

Read the labels and **value**s. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and **value**s. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: count+tag raises **Type**Error rather than silently converting tag. Changed condition: Change count to 10, leaving tag unchanged. Expected result: The two valid **output**s become 12 and 82.

Reason: Only the integer **variable** changes; the **string** still contains the character 8.

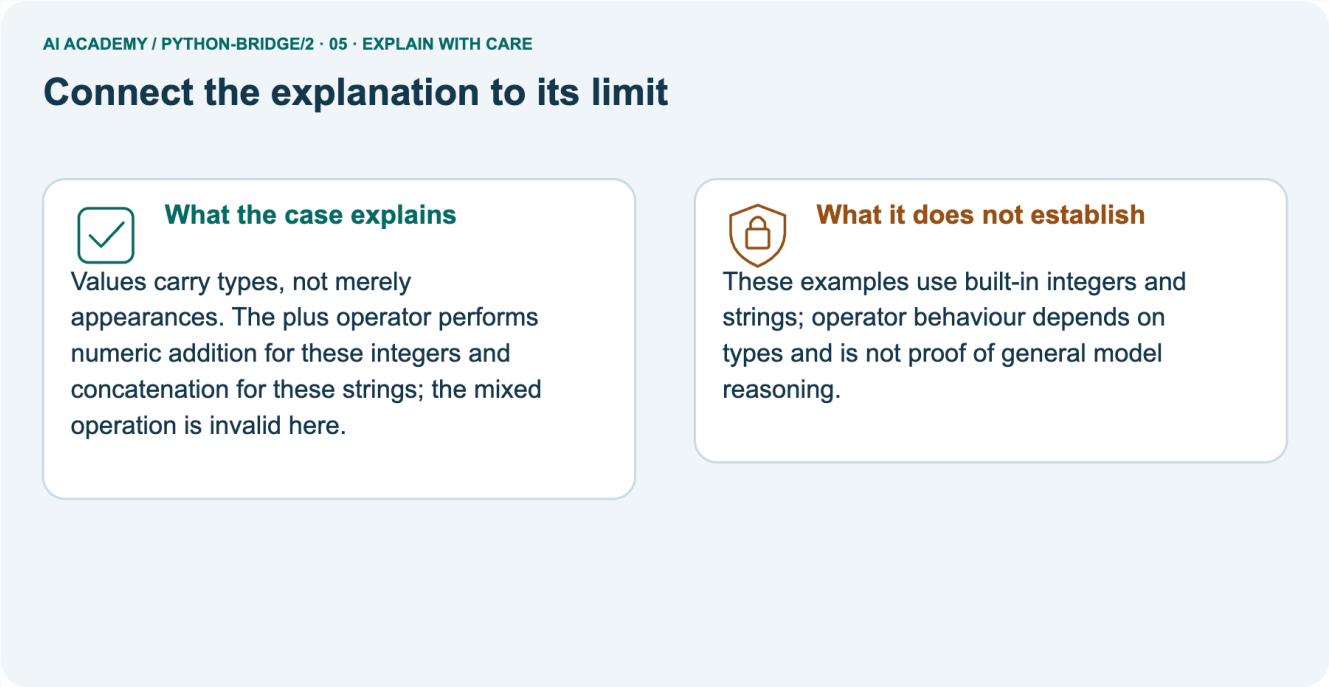
> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Only the integer **variable** changes; the **string** still contains the character 8.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Values carry **type**s, not merely appearances. The plus operator performs numeric addition for these integers and concatenation for these **string**s; the mixed operation is invalid here.

Limit: These examples use built-in integers and **string**s; operator behaviour depends on **type**s and is not proof of general model reasoning.

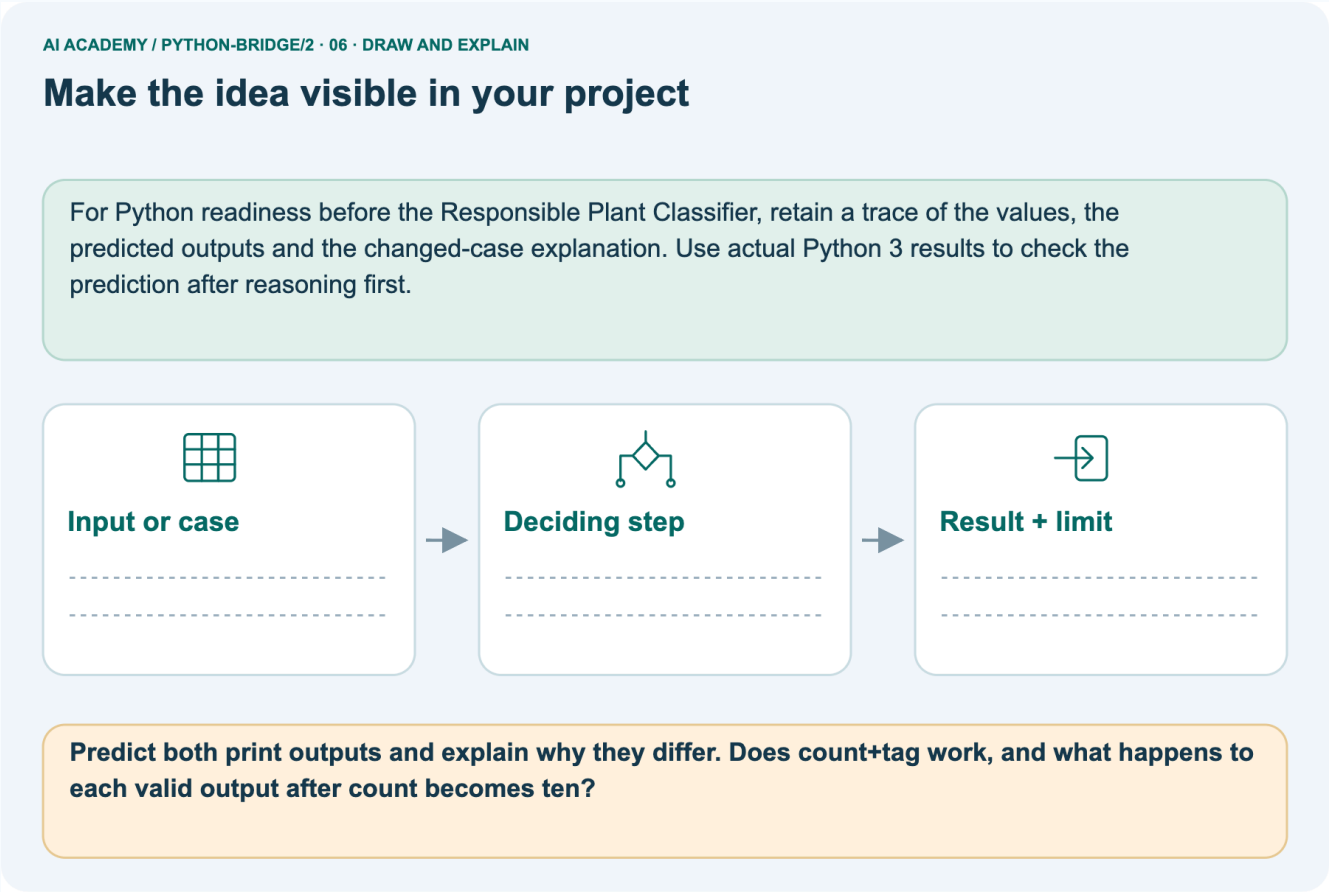
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: These examples use built-in integers and **string**s; operator behaviour depends on **type**s and is not proof of general model reasoning.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For Python readiness before the Responsible Plant Classifier, retain a trace of the **value**s, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Draw your **input**, deciding step, result and limit. Supported question: Predict both print **output**s and explain why they differ. Does count+tag work, and what happens to each valid **output** after count becomes ten?

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The **output**s are 10 and 82. The first adds integers; the second joins **string**s. count+tag raises **Type**Error. Changing count to 10 yields 12, while tag+"2" remains 82.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.